FRUIT TWIN GPT

Base-paper analysis and proposed research project plan

Project title:
A Digital Twin-Driven Multimodal Framework for Early Fruit Quality Prediction Using Vision-Language Models

Prepared from the uploaded project abstract and base paper

Base paper: Kim & Sun, IEEE Access, 2026, DOI 10.1109/ACCESS.2026.3697751

Research positioning
The base paper is a tabular cross-crop transfer-learning study. Your abstract proposes a broader early-stage, image-plus-environment digital-twin system. Extend the problem in measurable ways; do not claim that the base paper already uses ViT, GNN, or a vision-language model.

PART I — Understand the base paper

1. Identity and purpose

Title: Balanced Multi-Source Transfer Learning for Fruit Quality Prediction With Limited Data. **Authors:** Yongjun Kim and Eun Jung Sun. Published in IEEE Access in 2026.

The paper asks whether data from other crops can improve fruit-quality prediction when a target crop has few labelled records, without letting a much larger source dataset dominate. The authors call the failure caused by naïve pooling the **source overwhelming problem**: abundant source samples can drown out the target crop's signal.

2. Main goals and contributions

• **Identify source overwhelming:** show that naïvely pooling multiple source crops can cause negative transfer.

• **Propose BMTL:** Balanced Multi-Source Transfer Learning controls the source-target sample balance.

• **Estimate transferability:** train on each source separately and evaluate its MAE on held-out target validation samples.

• **Weight and sample sources:** give greater weight to sources with lower target-validation error and sample a controlled number of records.

• **Propose LOO-Select:** for extremely small target datasets (paper uses nT < 50), choose the best single source rather than combining all sources.

• **Evaluate broadly:** compare with target-only, naïve pooling, fine-tuning, CORAL, TCA, instance weighting and DANN-style transfer.

3. Datasets used and why

Dataset	What the paper reports	Purpose
Korean fruit-quality records (NIHHS)	2003–2024; apple 366, citrus 2,527, grape 473, pear 355; 3,721 quality measurements total	Real agricultural regression data with imbalanced crop sizes; each crop can be target while others act as sources.
Korea Meteorological Administration (KMA)	20,896 daily weather records: temperature, precipitation, sunshine and relative humidity	Weather affects fruit development and sugar accumulation; links growing conditions to quality.
UCI Wine Quality	White wine 4,898 source samples; red wine 1,599 target samples	Tests transfer on another tabular regression benchmark.
UCI Concrete Strength	1,030 samples; low-cement source 719, high-cement target 311	Tests transfer across subdomains with differing feature distributions.
UCI Energy Efficiency	768 samples; source 512 and target 256 based on compactness	Tests a high-domain-shift case where negative transfer is a concern.

Why these datasets? The Korean fruit records provide a real cross-crop setting with different sample counts. The three UCI benchmarks test whether the same imbalance problem appears beyond fruit data. The base paper says its Korean multi-crop dataset is available upon reasonable request; do not assume it is a direct public download.

4. Inputs, outputs and workflow

Input: tabular crop-quality and weather features from the target crop plus source crops. Features described include weight, Brix/sugar content, acidity, firmness, colour index, size, harvest timing, variety/region/year, growing-degree days and weather summaries.

Output: numeric fruit-quality regression predictions (the apple case study focuses on Brix/sugar content), evaluated with R^2 and error metrics. The paper does not generate fruit images or natural-language explanations.

• 1. Clean/standardize data; examine correlated features and use PCA retaining 95% variance in reported experiments.

• 2. Split target records into training and validation (the algorithm shows 80%/20%).

• 3. Train one model per source crop and measure MAE on target validation data.

• 4. Convert source errors to transferability scores using inverse MAE, then normalize them into weights.

• 5. Select a target/source balance ratio; the paper recommends trying values such as 0.3, 0.5 and 0.8.

• 6. Sample a controlled number of source records according to source weights.

• 7. Combine sampled source records with target training records and train the final regressor.

• 8. Compare with baselines on held-out data; do not use transfer if validation results are worse than target-only.

5. Models, algorithms and tools

Item	Role
BMTL	Main method: transferability estimation, balance-aware weighting and adaptive sampling.
LOO-Select	Selects one best source for very small target data; otherwise applies multi-source BMTL.
Ridge regression	Primary learner, $\alpha = 1.0$, chosen for interpretability and reproducibility.
Other regressors	Gradient Boosting Regressor, Random Forest and RBF-kernel SVR for comparison.
Baselines	Target-only, naïve pooling, fine-tuning, CORAL, TCA, instance weighting, DANN-style.
Preprocessing / evaluation	Z-score normalization, PCA, MMD domain distance, 10 random seeds, 80/20 splits, paired t-tests and Wilcoxon signed-rank tests.
Software detail	SciPy statistical tests are named. The paper does not specify a complete production application stack.

6. Main results reported

• Agricultural apple-target case: R^2 increases from 0.393 to 0.536 (+0.143 points; +36.4% relative).

• The paper reports up to +117.8% relative improvement for a tested setting with only 50 target training samples.

• BMTL is not universally best: on Energy, target-only R^2 is 0.553 and BMTL is 0.455. It avoids catastrophic failures of some other transfer methods but does not beat target-only in every case.

• When target data is extremely scarce ($nT < 50$), selecting a single source can be better than combining all sources.

7. Limitations and disadvantages

• **Regression focus:** conclusions do not automatically establish performance on classification tasks.

• **Needs labelled target validation data:** source ranking and balance tuning may be unreliable when target data is extremely small.

• **Theory assumes linear models:** deep neural architectures may require different analysis and balancing strategies.

• **Scaling cost:** it trains one model per source, so transferability estimation grows linearly with the number of sources.

• **Static balance ratio:** the current method does not dynamically schedule the ratio during training; adaptive/curriculum balancing is future work.

• **Transfer can hurt:** if sources are poorly matched, target data is already plentiful, or domain shift is high, BMTL may underperform target-only.

• **Difference from your abstract:** the paper is tabular and does not implement early flower/leaf image sensing, a plant-level digital twin, or a farmer-facing language explanation.

> The final point is a potential extension opportunity relative to your abstract, not proof that no prior research has ever addressed it.

PART II — Your proposed project

8. What your abstract proposes

Title: A Digital Twin-Driven Multimodal Framework for Early Fruit Quality Prediction Using Vision-Language Models.

Your abstract proposes predicting quality at individual-plant level before fruit has fully developed. It combines flower and leaf images with soil nutrients, weather, irrigation history, plant age and fruit variety. It names a Vision Transformer (ViT) for images, a Graph Neural Network (GNN) for relationships between the plant and its growing environment/management, and a Vision-Language Model (VLM) to explain predictions.

The listed outputs are fruit weight, sweetness (°Brix), firmness, shelf life and expected harvest date. Keywords: Digital Twin, Fruit Quality Prediction, Multimodal Learning, Precision Agriculture, Vision Transformer and Vision-Language Model.

9. Two main research objectives

Objective 1 — Multimodal early prediction: build and evaluate a model that fuses early-stage flower/leaf images with time-aligned weather, soil, irrigation, plant age and variety data to predict fruit-quality traits and harvest timing before harvest. Compare image-only, tabular-only and fused models.

Objective 2 — Explainable digital-twin decision support: create a plant-level digital-twin record tracking inputs, predictions, uncertainty and contributing factors, then generate a farmer-readable explanation grounded in model outputs. Evaluate explanation faithfulness and usefulness rather than treating fluent text as proof of correctness.

These are defensible extensions because the base paper focuses on balancing tabular source/target records for regression. It does not implement the image branch, multimodal fusion, plant-state digital twin or farmer-facing explanation workflow described in your abstract.

10. Required datasets and where to get them

The key requirement is **paired, time-aligned data**: images taken before harvest, environmental/management records, and a later measured outcome (for example Brix measured with a refractometer). Public image datasets alone generally do not contain all these linked fields.

Data needed	Potential source	Use / caution
Early flower/leaf images	Your own orchard/greenhouse collection; PlantVillage: https://data.mendeley.com/datasets/tywbtsjrjv/1	PlantVillage is mainly healthy/diseased leaf imagery, not paired early-stage fruit-quality labels. Useful for image pretraining, not as the sole predictor dataset.
Fruit/variety images	Fruits-360: https://github.com/fruits-360	Useful for fruit/variety recognition and pipeline tests. Isolated harvested fruit images do not substitute for plant-stage images or Brix labels.
Daily weather	NASA POWER Daily API: https://power.larc.nasa.gov/docs/services/api/temporal/daily/	Download available weather/solar variables by location and date; account for grid resolution and local microclimate.
Soil properties	FAO Soils Portal: https://www.fao.org/soils-portal/ and local laboratory soil tests	Use measured pH, nutrients and organic carbon where possible; broad maps are not plant-specific measurements.
Irrigation / farm management	Farmer logbook or controlled field experiment	Record dates, irrigation amount, fertilizer, pruning, planting date, variety, plant age and plot ID.
Ground-truth quality	Measure at harvest: Brix, weight, firmness, acidity, actual harvest date; shelf-life observations if feasible	Essential supervised labels. Use a consistent instrument and measurement protocol.

Recommended scope: start with one crop and one or two varieties available locally. Collect repeated images at defined growth stages and assign a unique plant_id/plot_id. Link each plant to harvest measurements. A small, well-labelled paired dataset is more useful for this question than a large unrelated fruit-image collection.

11. Suggested data schema

File/table	Example fields
plant_registry.csv	plant_id, plot_id, crop, variety, planting_date, location, soil_test_id
images/	plant_id, capture_date, growth_stage, image_type (flower/leaf/fruit), image_path
weather_daily.csv	location_id, date, min_temp, max_temp, mean_temp, rainfall, humidity, sunshine
management.csv	plant_id/plot_id, date, irrigation_amount, fertilizer_type/amount, treatment

File/table	Example fields
harvest_labels.csv	plant_id, harvest_date, weight_g, brix, firmness, acidity, shelf_life_days

Avoid data leakage: early-prediction models must not use images captured after the prediction date. Split by plant, plot and preferably season, not by image alone, to prevent near-duplicate views of the same plant leaking across train/test.

12. Models and technologies to use

Component	Recommendation	Why
Image encoder	Pretrained ViT or CNN baseline using PyTorch	Extract image features; fine-tune only if data volume supports it.
Tabular model	CatBoost / Random Forest / Gradient Boosting and an MLP baseline	Strong structured-data baselines.
Fusion	Concatenate image embeddings and normalized tabular features; compare late fusion or gated/attention fusion	Gives a measurable modality-fusion contribution.
Digital twin	Plant-level state record and prediction history	A useful software representation does not require a complex 3D simulation.
GNN (optional)	PyTorch Geometric, only with real graph relationships	Use if plants, plots, weather stations and management records form meaningful nodes/edges; compare with non-graph baselines.
Explainability	SHAP for tabular factors; image saliency; uncertainty intervals	Evidence for explanations; saliency is not proof of causality.
VLM explanation	Provide the structured prediction, uncertainty and verified feature attributions as input	Language layer should explain evidence, not invent agronomic causes.
Full stack	React dashboard; Python FastAPI ML service; optional Spring Boot API and MySQL	The app stores plants, measurements, predictions and history. Spring Boot can call Python inference over REST.
Experiment tooling	pandas, scikit-learn, PyTorch, experiment logs/MLflow	Reproducible preprocessing and model comparison.

13. End-to-end project workflow

• 1. Register plant/plot, crop variety, location, planting date and soil test.

• 2. Capture flower/leaf images at predefined early stages; save plant_id, date and growth stage.

• 3. Retrieve weather by location/date and append actual irrigation/management records.

• 4. Clean data, align time windows, standardize units, encode categories and document missingness.

• 5. Train three baselines: tabular-only, image-only and fused multimodal model.

• 6. Predict Brix, weight, firmness and harvest date; include shelf life only if labels are measured consistently.

• 7. Evaluate on held-out plants/plots/seasons with R^2, MAE and RMSE; measure harvest-date error in days.

• 8. If multi-crop/variety data supports it, compare target-only, naïve pooling, BMTL and your fusion method.

• 9. Produce SHAP/tabular contributions, visual evidence and uncertainty; pass verified information to the explanation generator.

• 10. Store input snapshots, model version, predictions, explanation and later measured outcomes in the digital twin.

• 11. Dashboard displays predicted quality, expected harvest window, uncertainty, contributing factors and prediction-vs-actual history.

14. Inputs and outputs

Inputs	Outputs
Flower/leaf images and capture date	Predicted Brix, fruit weight, firmness and harvest date—only for targets with training labels.
Crop/variety, plant age, soil/site data	Confidence or uncertainty interval if properly calibrated.
Daily/seasonal weather and growing-degree days	Important contributing factors and cautious visual evidence summaries.

Inputs	Outputs
Irrigation, fertilizer and management history	Farmer-readable explanation tied to model evidence and limitations.
Previous predictions and harvest measurements	Digital-twin timeline and prediction-versus-actual tracking.

15. Experiments for the conference paper

• **A. Modality ablation:** tabular-only vs image-only vs fused model.

• **B. Model comparison:** Random Forest/Gradient Boosting vs ViT+MLP fusion; GNN only if graph data exists.

• **C. Early-stage test:** predict at multiple growth stages without using future information.

• **D. Transfer baseline:** target-only vs naïve pooling vs BMTL where multiple crops/varieties have enough labels.

• **E. Explanation evaluation:** check agreement with model attributions and whether users understand uncertainty; do not claim explanations are causal.

• **Metrics:** MAE, RMSE, R^2; harvest-date MAE in days; uncertainty calibration/coverage; modality ablation; runtime.

• **Validation:** group splits by plant/plot/season and, if possible, test on a different farm or season.

16. Risks and scope decisions

• The listed public datasets do not by themselves supply the complete paired early-stage image + weather + management + Brix dataset. Field collection or a data-sharing agreement is likely necessary.

• Shelf-life prediction requires a defined storage protocol and repeated observations; keep it optional if you cannot collect it reliably.

• A GNN and VLM add complexity. First build the multimodal baseline and reliable dataset.

• The base paper's Korean dataset is available upon request according to the paper. Contact the authors if exact replication is needed.

• Do not claim 'first ever' without a current systematic literature review. Frame novelty around your data alignment, early-stage setting, fusion comparisons and evaluated explanation workflow.

17. Recommended final project definition

Core project: a plant-level digital twin that combines early flower/leaf images with weather, soil and management records to predict measured fruit-quality traits before harvest, with evidence-grounded explanations.

Achievable first version: one crop, two growth stages, Brix + weight + harvest date, image encoder + tabular model + fusion, SHAP-based evidence, React dashboard and Python inference API. Add GNN/VLM only after the baseline works and the data supports them.

Two objectives to put in the proposal: (1) design and evaluate early-stage multimodal fusion for fruit-quality and harvest-time prediction; (2) develop a digital-twin decision-support layer that tracks plant state and produces interpretable, evidence-grounded explanations with uncertainty.

References and dataset links

• Kim, Y. & Sun, E. J. (2026). “Balanced Multi-Source Transfer Learning for Fruit Quality Prediction With Limited Data.” IEEE Access. DOI: https://doi.org/10.1109/ACCESS.2026.3697751 (uploaded base paper).

• Fruits-360: https://github.com/fruits-360 — fruit/variety images; not early-stage quality labels.

• PlantVillage dataset: https://data.mendeley.com/datasets/tywbtsjrjv/1 — leaf-health imagery; not paired fruit-quality labels.

• NASA POWER Daily API: https://power.larc.nasa.gov/docs/services/api/temporal/daily/ — daily meteorological/solar data.

• FAO Soils Portal: https://www.fao.org/soils-portal/ — soil information and legacy maps; local measurements are preferable for plant-level ground truth.

Source boundary: Base-paper facts and its stated limitations are summarised from the uploaded IEEE Access PDF. Project objectives, architecture, data plan and experiments are recommendations based on your uploaded abstract and research-planning judgement. Check dataset access and licenses before publication.